

Vehicle Owner's Manual — General Reference

A general-purpose reference covering the core systems found in most passenger vehicles. This document is written for the vehicle health intelligence knowledge base and should be read alongside your vehicle-specific manual, which takes precedence for any conflicting torque values, fluid specifications, or service intervals.

1. Engine System

The internal combustion engine converts fuel into mechanical energy through a repeating cycle of intake, compression, combustion, and exhaust. Engine health is primarily monitored through four parameters: coolant/oil temperature, oil pressure, rotational speed (RPM), and vibration.

Temperature

Normal operating temperature for most engines falls between 80 and 95 degrees Celsius once warmed up. Temperatures above 100 degrees Celsius indicate the cooling system is not adequately dissipating heat and can lead to warped cylinder heads, blown gaskets, or seized components if sustained. Common causes of overheating include low coolant level, a failed thermostat, a clogged radiator, a failing water pump, or a slipping/broken serpentine belt driving the water pump.

Oil Pressure

Oil pressure lubricates moving parts and carries heat away from the engine's internals. Low oil pressure at idle or under load is one of the most serious warning signs a vehicle can give — continuing to drive with low oil pressure risks catastrophic bearing and camshaft damage within minutes. If an oil pressure warning light illuminates, stop the vehicle as soon as it is safe to do so and check the oil level before continuing.

Vibration and Rough Running

Excessive engine vibration usually points to one of: a cylinder misfire (failing spark plug, ignition coil, or fuel injector), worn or broken engine mounts, an imbalanced accessory (alternator, power steering pump), or a failing timing component. Vibration that worsens with RPM often indicates a rotating imbalance; vibration that is worst at idle and smooths out under load more often points to a misfire.

2. Battery & Electrical System

The 12V battery (or auxiliary battery in hybrid/EV vehicles) starts the engine and powers accessories when the alternator/DC-DC converter is not supplying enough current on its own. Key health indicators are terminal voltage, current draw, and internal temperature.

Voltage

A healthy, fully-charged 12V lead-acid battery reads approximately 12.6V at rest. Readings between 11.8V and 12.4V indicate a partially discharged battery that should be recharged and load tested. Readings below 11.8V typically indicate a battery that cannot reliably start the engine and should be tested for replacement. With the engine running, alternator output should raise system voltage to roughly 13.8–14.7V; readings outside this band suggest a charging system fault rather than a battery fault.

Current Draw and Parasitic Drain

Elevated current draw with the engine off ("parasitic drain") will slowly discharge even a healthy battery. Common causes include a interior light left on, a faulty relay stuck closed, or a control module that has failed to enter sleep mode. Sustained high current draw while driving, beyond what accessories should require, can indicate a short circuit and warrants immediate inspection.

Temperature

Battery chemistry is temperature-sensitive: cold reduces available cranking current, while sustained heat above roughly 50 degrees Celsius accelerates internal corrosion and shortens service life. Batteries mounted near the engine bay in hot climates typically have shorter lifespans than those in cooler, better-ventilated locations.

3. Fuel System

The fuel system stores, filters, pressurizes, and delivers fuel to the engine. A healthy fuel system maintains consistent pressure and delivers clean, correctly-metered fuel to each cylinder.

Common fuel system health indicators include fuel trim values (how much the engine computer is correcting the commanded air-fuel ratio), fuel pressure at the rail, and injector pulse width consistency across cylinders. Symptoms of a degrading fuel system include hesitation under acceleration, difficulty starting (especially when hot), reduced fuel economy, and, in advanced cases, misfires under load. Root causes range from a clogged fuel filter or dirty injectors to a weakening fuel pump or a failing fuel pressure regulator.

4. Braking System

Brakes convert kinetic energy into heat through friction between pads/shoes and rotors/drums. Because this is the vehicle's primary safety system, brake health should never be deferred.

- Pad/shoe wear: friction material below roughly 3mm should be replaced; most vehicles have an audible wear indicator that produces a high-pitched squeal once this threshold is reached.
- Brake fluid: absorbs moisture over time (it is hygroscopic), which lowers its boiling point and can lead to a spongy pedal under hard braking. Most manufacturers recommend a fluid flush every two years regardless of mileage.
- Rotor condition: warping (usually from repeated hard stops without adequate cooling) causes pedal pulsation; scoring (from worn-out pads contacting bare rotor) reduces stopping power and should prompt resurfacing or replacement.
- ABS/traction system: a dedicated warning light indicates a fault in the anti-lock braking system; the vehicle retains normal (non-ABS) braking but should be inspected promptly, as ABS is a significant safety feature in emergency stops.

5. Dashboard Warning Lights — General Guide

Warning light color is a rough severity indicator across most manufacturers, though the exact meaning of any given icon should always be confirmed against the vehicle-specific manual.

Color	General Meaning	Recommended Response
Red	Safety-critical or system-stopping fault (e.g. oil pressure, brake system, coolant temperature)	Stop driving as soon as safely possible; do not continue the trip unless advised by a qualified technician
Yellow / Amber	Non-critical fault requiring attention (e.g. check engine, tire pressure)	Service soon; vehicle is generally safe to drive short distances
Green / Blue	Informational — a system is active (e.g. headlights, cruise control)	No action needed